Mohammad Shaheer

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Education

Carnegie Mellon University **Pittsburgh, PA**
Masters of Science in Computer Science May 2027
Carnegie Mellon University **Doha, Qatar**
Masters of Science in Computer Science | Concentration in Algorithms & Complexity May 2023
University Honors & College Honors
Relevant Coursework: Planning and Decision Making in Robotics, Introduction to Robot Learning

Skills

Programming Languages: Python, C/C++/C#, Java, SML, Typescript/Javascript
Frameworks & Databases: Django REST Framework/Django, .NET, Flutter, NextJS, MongoDB, PostgreSQL
Cloud & Tools: GCP, AWS, Azure, Heroku, Docker, Kubernetes, Isaac Lab/Isaac Sim, ROS, Gazebo, rviz

Professional Experience

AirLab - Carnegie Mellon University | Robotics Researcher | Pittsburgh, PA May. 2026 - Present
- Learning multi-modal abstract world models for long-horizon task and motion planning via coding agents
- Using RGB videos and motion capture data to perform real-to-sim in IsaacSim in order to learn low-level control policies
- Defining appropriate RL based environments in IsaacLab to train and test control policies and task and planning approaches

Avey | Full Stack Software Engineer | Doha, Qatar Dec. 2023 - May. 2026
- Developed Coder, an AI-powered insurance claim auditing platform to **automate error detection and reduce claim rejections**; deployed in 2 local hospitals supporting 4 insurance companies
- Engineered a token escrow system to prevent overcharging; **reducing monthly billing errors from ~30% to ~3%**
- Integrated with cloud-deployed AI models for medical adjudication and surfaced results on the web app **reducing claim rejection by ~10%**
- Replaced ad hoc Slack-based approval processes with a centralized, audited system, **cutting average approval time from days to hours across ~100+ monthly requests**
- Optimized Django ORM queries on a high-traffic list endpoint, **reducing response time from ~900ms to ~45ms** by profiling query execution and eliminating N+1 patterns

Snoonu | Software Engineer | Doha, Qatar May 2023 - Dec 2023
- Conducted EDA on marketing campaign data to assess feature suitability for demand prediction models
- **Improved testing accuracy of demand forecasting model by ~17%** through hyperparameter tuning
- Deployed a real-time driver tracking system communicating driver status updates to the merchant portal periodically via SignalR and RabbitMQ **reducing merchant calls to operations team by 37%**

Projects

Distributed File System | Java Carnegie Mellon University | Nov. 2022
- Built a distributed file system using custom RMI with multi-threaded Skeleton/Stub architecture, **enabling remote file operations across multiple storage servers**
- Implemented a metadata management system on a centralized naming server, coordinating storage server registration, duplicate pruning, and synchronized create/delete operations across nodes
- Implemented hierarchical read-write locking with FIFO-ordered queues and root-to-leaf path locking, **ensuring consistency for concurrent multi-client reads and writes while preventing deadlocks**
- Engineered a dynamic replication engine that auto-scaled file replicas based on request load using a linear coarse-grained policy, with invalidation-based write propagation **to keep replicas consistent**

Moon Ranger | C/C++ Carnegie Mellon University | Jan. 2022 - Sept. 2022
- Worked as a software engineer on the MoonRanger project to develop an autonomous rover for lunar pole water search
- Extended NASA's Core Flight System software bus library to support local data storage, **enabling autonomous rover operations without constant communication links**
- Optimized rover perception pipeline by reducing background noise in disparity maps, **improving navigation accuracy** for lunar terrain analysis